

9.1 Database Fundamentals

A **database** is a collection of data that is multidimensional, meaning:

- data items are linked to each other
- information can be accessed from multiple perspectives

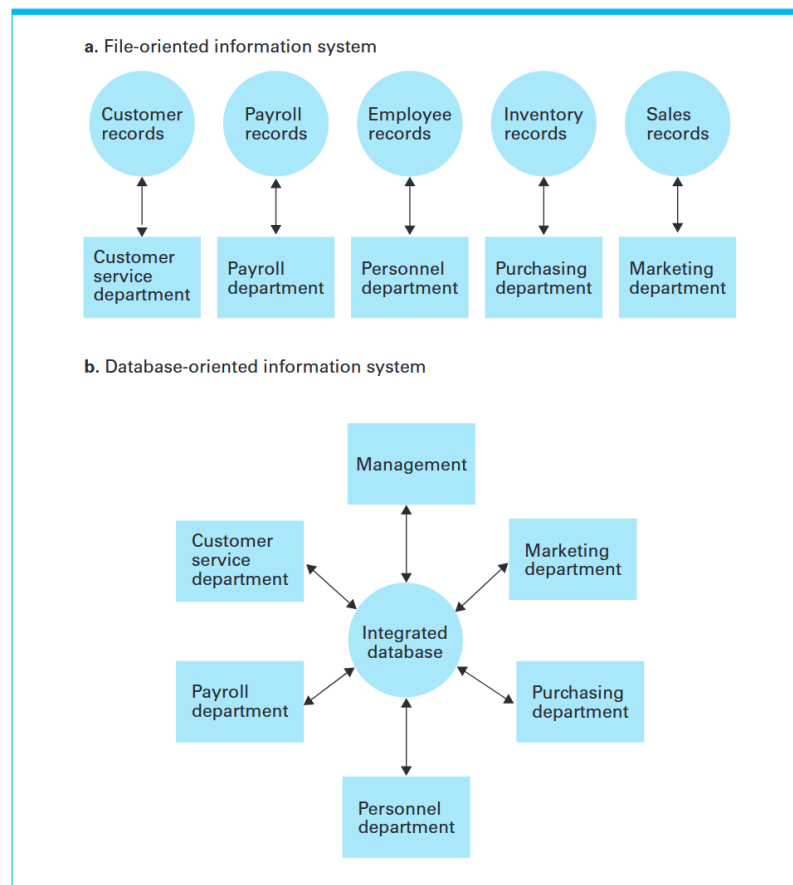
flat file

A flat file is a traditional file system that is:

- one-dimensional
- provides data from a single point of view

The Significance of Database Systems

Figure 9.1 A file versus a database organization



The Role of Schemas

A **schema** is a description of the entire database structure.

It defines:

- what data exists
- how data is organized
- how data is linked

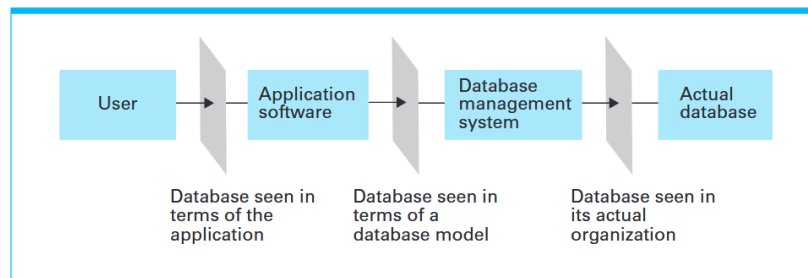
A **subschemas** is a description of only the portion of the database relevant to a specific user.

Database Management Systems

A **database management system (DBMS)** is: software that manages and controls access to a database, performing all data manipulation operations such as storing, retrieving, and updating data.

用户 → 应用 → DBMS → 数据库 → 返回结果

Figure 9.2 The conceptual layers of a database implementation



A **distributed database** is a database whose data is stored across multiple machines connected by a network.

Data independence is the ability to change the structure of the database without modifying the application software.

database model

A **database model** is a conceptual representation of how data in a database is organized and accessed.

1. Identify two departments in a manufacturing plant that would have different uses for the same or similar inventory information. Then, describe how the subschema for the two departments might differ.

教务处能查学生导师是谁，但不能看老师的工资 / 工作经历。

工资部门能查老师工资 / 修改工资，但不能查这个老师带哪些学生。

2. What is the purpose of a database model?

The purpose of a database model is:

- to provide a conceptual view of the database
- to allow users to interact with data in a simple and meaningful way
- to hide the complexities of the database's internal structure

3. Summarize the roles of the application software and a DBMS.

The application software:

interacts with the user

determines what actions the user requests

The DBMS:

performs all database operations

stores, retrieves, and updates data

enforces access control based on schemas and subschemas

9.2 The Relational Model

In the **relational model**, this multidimensional structure is represented as a collection of **tables**.

A **relation** is a table consisting of rows and columns.

Tuple 元组: a row in a relation, representing a single record.

Attributes 属性: a column in a relation, representing a property of the data.

Issues of Relational Design

Figure 9.3 A relation containing employee information

Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
⋮	⋮	⋮	⋮

Problem 1: Redundancy (冗余)

以前：一个人一行 现在：一次工作一行

Problem 2: Inefficiency (低效)

浪费空间 修改困难 (要改多处)

Problem 3: Deletion Anomaly

如果某个工作只有一个人做：删除这个人 → 工作信息也没了

Figure 9.4 A relation containing redundancy

Empl Id	Name	Address	SSN	Job Id	Job Title	Skill Code	Dept	Start Date	Term Date
25X15	Joe E. Baker	33 Nowhere St.	111223333	F5	Floor manager	FM3	Sales	9-1-2009	9-30-2010
25X15	Joe E. Baker	33 Nowhere St.	111223333	D7	Dept. head	K2	Sales	10-1-2010	*
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999	F5	Floor manager	FM3	Sales	10-1-2009	*
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555	S25X	Secretary	T5	Personnel	3-1-1999	4-30-2010
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555	S26Z	Secretary	T6	Accounting	5-1-2010	*
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Figure 9.5 An employee database consisting of three relations

EMPLOYEE relation			
Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555
⋮	⋮	⋮	⋮
JOB relation			
Job Id	Job Title	Skill Code	Dept
S25X	Secretary	T5	Personnel
S26Z	Secretary	T6	Accounting
F5	Floor manager	FM3	Sales
⋮	⋮	⋮	⋮
ASSIGNMENT relation			
Empl Id	Job Id	Start Date	Term Date
23Y34	S25X	3-1-1999	4-30-2010
34Y70	F5	10-1-2009	*
23Y34	S26Z	5-1-2010	*
⋮	⋮	⋮	⋮

Figure 9.6 Finding the departments in which employee 23Y34 has worked

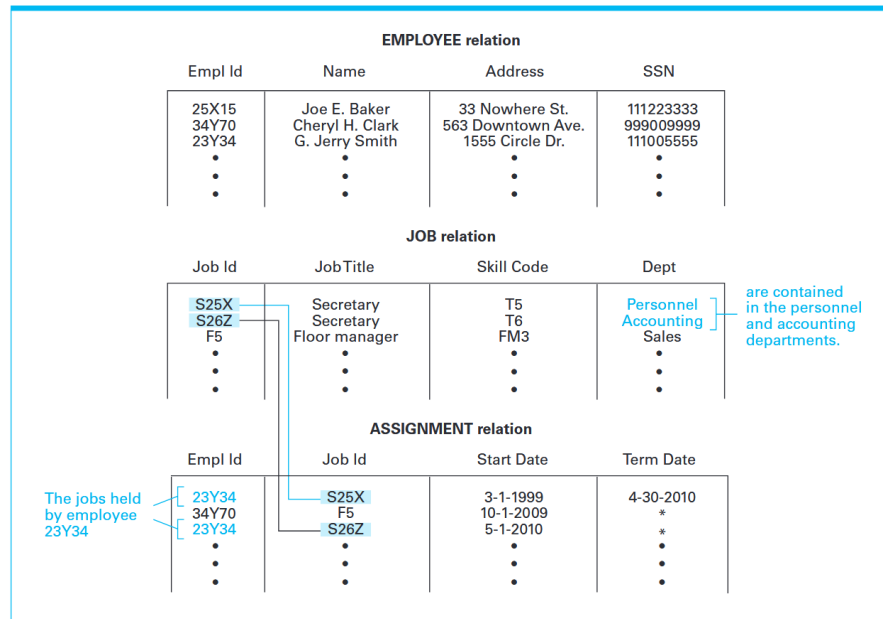
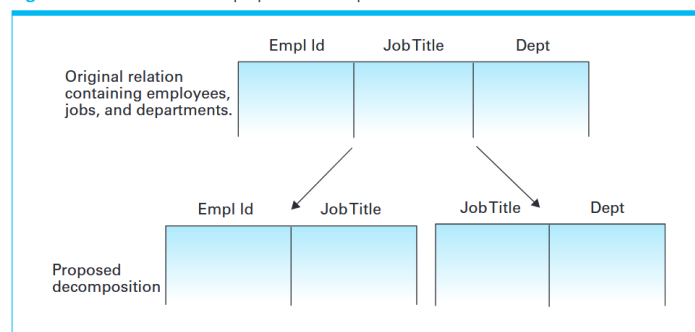


Figure 9.7 A relation and a proposed decomposition



lossless decomposition (nonloss decomposition)

Relational Operations

Figure 9.8 The SELECT operation

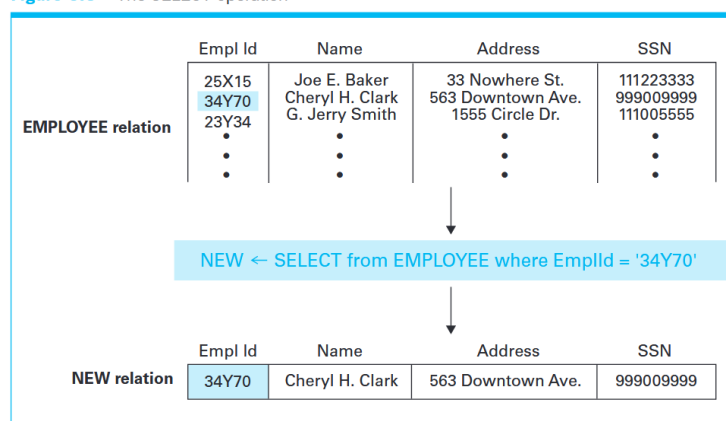


Figure 9.9 The PROJECT operation

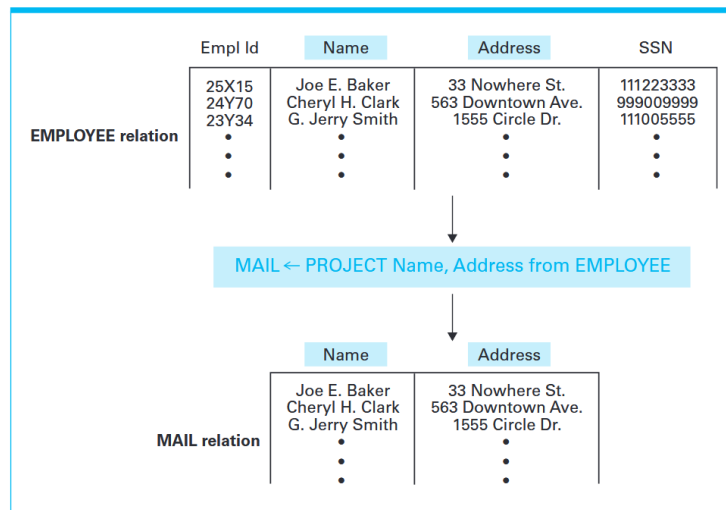
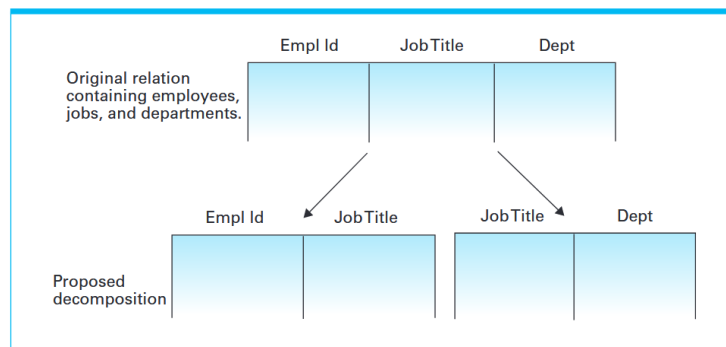


Figure 9.7 A relation and a proposed decomposition



知道员工是 Manager

但 A 部门和 B 部门都有 manger 这个 jobtitle

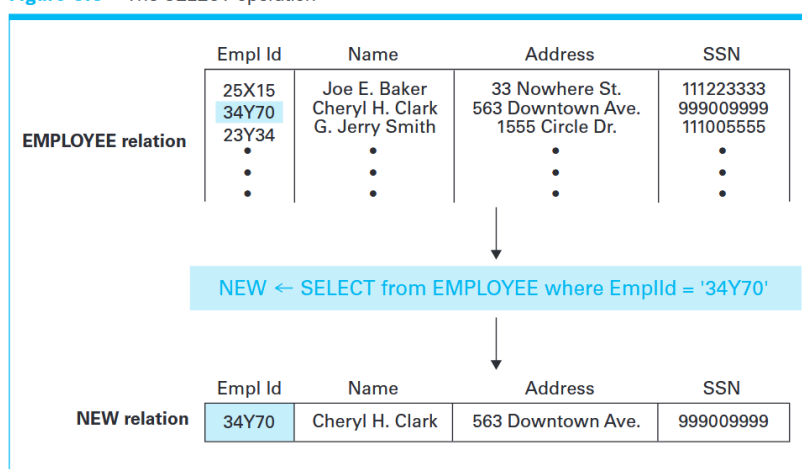
lossless decomposition (nonloss decomposition) (无损分解)

the original information can be fully reconstructed from the smaller relations.

Relational Operations

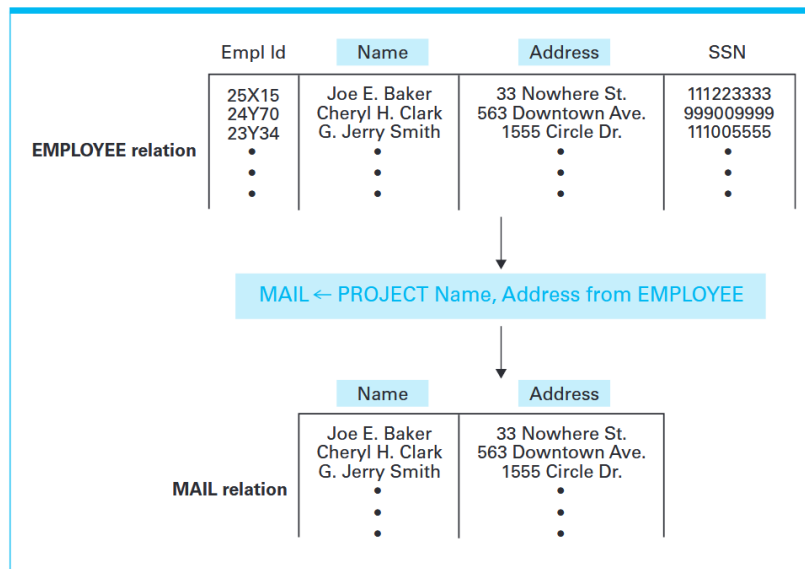
SELECT → choose rows

Figure 9.8 The SELECT operation



PROJECT → choose columns

Figure 9.9 The PROJECT operation



JOIN → combine relations

Figure 9.10 The JOIN operation

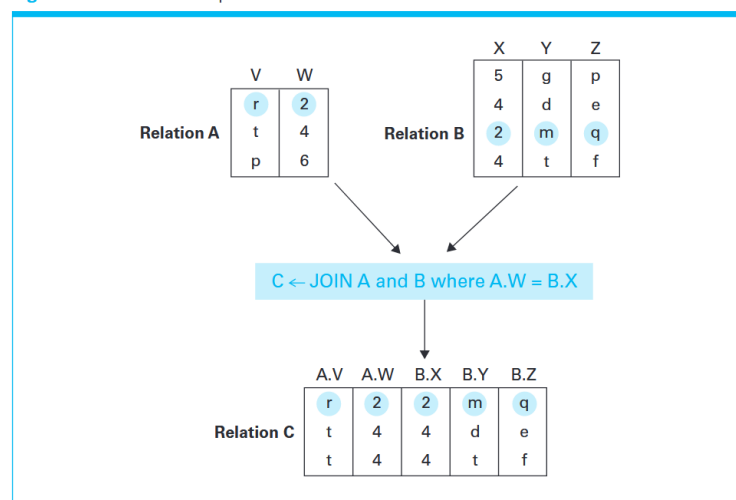


Figure 9.11 Another example of the JOIN operation

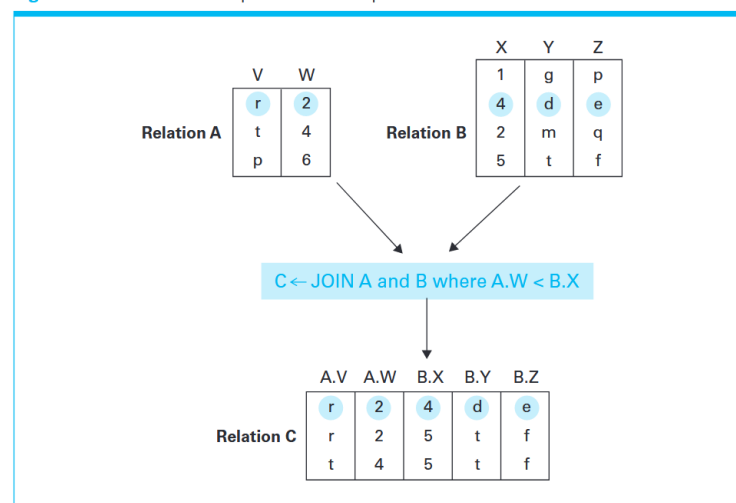
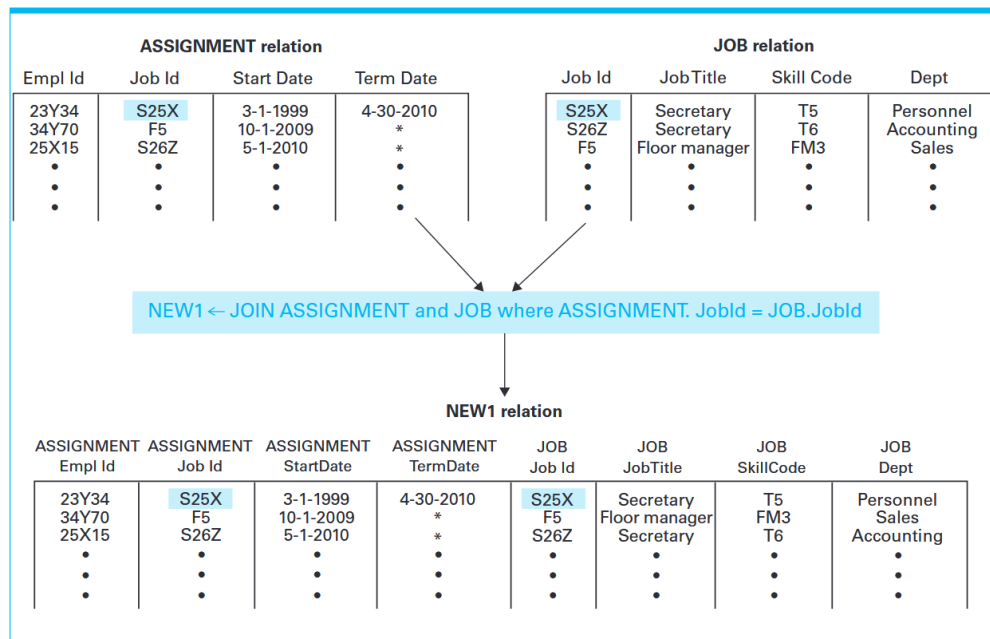


Figure 9.12 An application of the JOIN operation



SQL

```
SELECT EmplId, Dept
FROM Assignment, Job
WHERE Assignment.JobId = Job.JobId
AND Assignment.TermDate = '*';
```

PROJECT

```
SELECT Name, Address FROM Employee;
```

SELECT

```
SELECT EmplId, Name, Address, SSNum
FROM Employee
WHERE Name = 'Cheryl H. Clark';
```

JOIN

```
SELECT Employee.Name, Assignment.StartDate
FROM Employee, Assignment
WHERE Employee.EmplId = Assignment.EmplId;
```

INSERT

```
INSERT INTO Employee
VALUES ('42Z12', 'Sue A. Burt', '33 Fair St.', '444661111');
```

DELETE

```
DELETE FROM Employee
WHERE Name = 'G. Jerry Smith';
```

UPDATE

```
UPDATE Employee
SET Address = '1812 Napoleon Ave.'
WHERE Name = 'Joe E. Baker';
```

1. Answer the following questions based on the partial information given in the EMPLOYEE, JOB, and ASSIGNMENT relations in Figure 9.5:

Figure 9.5 An employee database consisting of three relations

EMPLOYEE relation			
Empl Id	Name	Address	SSN
25X15	Joe E. Baker	33 Nowhere St.	111223333
34Y70	Cheryl H. Clark	563 Downtown Ave.	999009999
23Y34	G. Jerry Smith	1555 Circle Dr.	111005555

JOB relation			
Job Id	JobTitle	Skill Code	Dept
S25X	Secretary	T5	Personnel
S26Z	Secretary	T6	Accounting
F5	Floor manager	FM3	Sales
•	•	•	•
•	•	•	•
•	•	•	•

ASSIGNMENT relation			
Empl Id	Job Id	Start Date	Term Date
23Y34	S25X	3-1-1999	4-30-2010
34Y70	F5	10-1-2009	*
23Y34	S26Z	5-1-2010	*
•	•	•	•
•	•	•	•
•	•	•	•

a. Who is the secretary in the accounting department with experience in the personnel department?

G. Jerry Smith

b. Who is the floor manager in the sales department?

Cheryl H. Clark

c. What job does G. Jerry Smith currently hold?

Secretary

2. Based on the EMPLOYEE, JOB, and ASSIGNMENT relations presented in Figure 9.5, write a sequence of relational operations to obtain a list of all job titles within the personnel department.

π JobTitle (σ Dept = 'Personnel' (JOB))

3. Based on the EMPLOYEE, JOB, and ASSIGNMENT relations presented in Figure 9.5, write a sequence of relational operations to obtain a list of employee names along with the employees' departments.

π Name, Dept ((EMPLOYEE $\bowtie$ ASSIGNMENT) $\bowtie$ JOB)

4. Convert your answers to questions 2 and 3 into SQL.

```
SELECT JobTitle
FROM JOB
WHERE Dept = 'Personnel'
```

```
SELECT E.Name, J.Dept
FROM EMPLOYEE E
JOIN ASSIGNMENT A ON E.EmplId = A.EmplId
```



```
JOIN JOB J ON A.JobId = J.JobId;
```

5. How does the relational model provide for data independence?

It hides how data is physically stored, so changes in storage do not affect how users access the data.

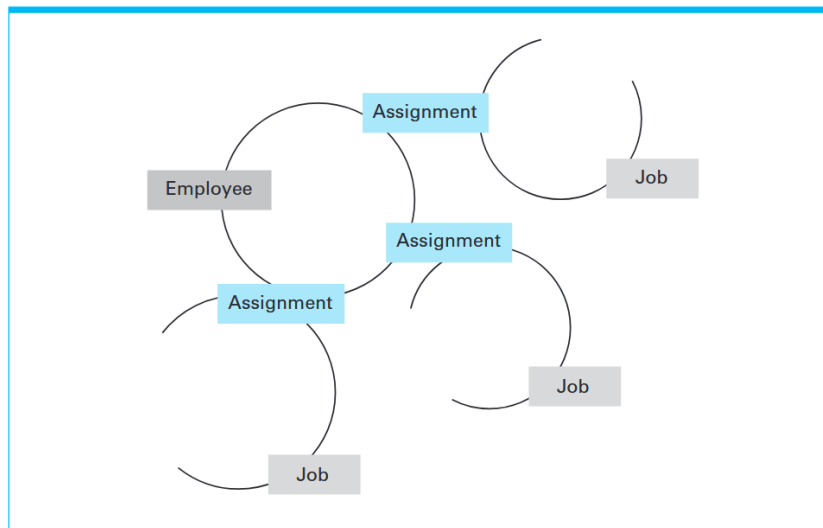
6. How are the different relations in a relational database tied together?

A primary key uniquely identifies a record in one relation, and a foreign key in another relation refers to it, linking the tables together.

9.3 Object-Oriented Databases

用“对象 (object) + 类 (class) + 链接 (link)”来组织数据

Figure 9.13 The associations between objects in an object-oriented database



Transient objects: disappear after program execution 临时的

Persistent objects: remain stored in the database 持久的

方面 Relational Object-Oriented

数据结构 表 对象

查询方式 JOIN follow links

数据类型 固定 灵活

行为 无 有 methods

结构 分离 封装

1. What methods would be contained in an instance of an object from the Assignment class in the employee database discussed in this section?

reporting the assignment details (such as start date and termination date),
updating the assignment information,
retrieving the associated job or employee.

2. What is a persistent object?

A persistent object is an object that remains stored in the database even after the program that created it terminates.

3. Identify some classes, as well as some of their internal characteristics, that can be used in an object-oriented database dealing with a warehouse inventory.

Product: attributes such as product ID, name, quantity, and price

Supplier: attributes such as supplier ID, name, and contact information

Shipment: attributes such as shipment date, product list, and quantity

4. Identify an advantage that an object-oriented database can have over a relational database.

An advantage of an object-oriented database is that it supports encapsulation, allowing data and the operations on that data to be combined within objects.

9.4 Maintaining Database Integrity

operations are completed correctly

interactions between operations do not corrupt data

The Commit/Rollback Protocol

The DBMS maintains a **log**:

records all actions before they are executed

stored in nonvolatile storage

commit point: the moment when all steps of a transaction are recorded in the log.

At this point, the DBMS guarantees that the transaction will be completed.

roll back: to undo all actions of a transaction

cascading rollback: rolling back one transaction forces other dependent transactions to be rolled back as well.级联回滚

A 修改数据 (但没 commit)

B 用了 A 的数据

A 回滚 → B 也必须回滚

开始事务

写日志

到达 commit → 成功

出问题 → rollback

Locking

Two Classic Problems

-**incorrect summary problem** 汇总错误问题

a transaction calculates a summary while another transaction is updating data.

-**lost update problem** 更新丢失问题

two transactions update the same data based on the same old value.

locking protocol: controls access to data items using locks

Two Types of Locks

- **Shared Locks (共享锁)**

allows multiple transactions to read the same data

多人可以读 不能改

- **Exclusive Locks (排他锁)**

allows only one transaction to access and modify the data

只能一个人用 可以修改

wound-wait protocol 伤-等待协议

older transactions preempt younger ones

younger transactions wait

1. What is the difference between a transaction that has reached its commit point and one that has not?

A transaction that has reached its **commit point** is guaranteed to be completed, whereas a transaction that has not reached its commit point may still be rolled back.

2. How could a DBMS guard against extensive cascading rollback?

A DBMS can guard against extensive cascading rollback by preventing transactions from

accessing data that has been modified by uncommitted transactions.

A 修改数据 (但没 commit)

B 用了 A 的数据

A 回滚 → B 也必须回滚

如果一开始就规定:

B 不能读 A 未提交的数据

那么:

B 不会依赖 A

A 回滚 → 不影响 B

级联被阻断

3. Show how the uncontrolled interweaving of two transactions, one deducting \$100 from an account and the other deducting \$200 from the same account, could produce final balances of \$100, \$200, and \$300, assuming that the initial balance is \$400.

Transaction A (扣 100)

1. read balance
2. subtract 100
3. write back

Transaction B (扣 200)

1. read balance
2. subtract 200
3. write back

Case 1: Final balance = 100

A completes, then B executes:

- A reads 400
- A writes 300
- B reads 300
- B writes 100

Final balance = **100**

Case 2: Final balance = 200 (错误: lost update)

- A reads 400
- B reads 400
- A writes 300
- B writes 200

Final balance = 200

Case 3: Final balance = 300 (错误: lost update)

- A reads 400
- B reads 400
- B writes 200
- A writes 300

Final balance = **300**

4. a. Summarize the possible results of a transaction requesting shared access to an item in a database.

the item is unlocked or already has a shared lock -> success

the item has an exclusive lock -> fail

b. Summarize the possible results of a transaction requesting exclusive access to an item in a database.

only if the item has no lock -> success

otherwise, the request is denied

5. Describe a sequence of events that would lead to deadlock among transactions performing operations on a database system.

Transaction A obtains an exclusive lock on item X.

Transaction B obtains an exclusive lock on item Y.

Transaction A then requests access to item Y and must wait.

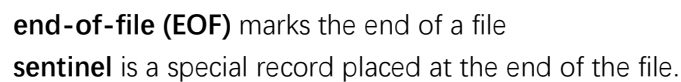
Transaction B then requests access to item X and must wait.

6. Describe how the deadlock in your answer to question 5 could be broken. Would your solution require use of the database management system's log? Explain your answer.

Yes. Roll back the younger transaction to allow the older one to proceed. The log is needed to undo the rolled-back transaction.

Sequential Files

Figure 9.14 The structure of a simple employee file implemented as a text file



```
def MergeFiles (InputFileA, InputFileB, OutputFile):
    if (both input files at EOF):
        Stop, with OutputFile empty.
    if (InputFileA not at EOF):
        Declare its first record to be its current record.
    if (InputFileB not at EOF):
        Declare its first record to be its current record.
    while (neither input file at EOF):
        Put the current record with the "smaller" key field
        value in OutputFile.
        if (that current record is the
            last record in its corresponding input file):
            Declare that input file to be at EOF.
    else:
        Declare the next record in that input file
        to be the file's current record.
    Starting with the current record in the input file that is not at
    EOF, copy the remaining records to OutputFile.
```

The diagram illustrates the merge sort algorithm for the array [A, B, C, D, E, F]. It shows the recursive splitting of the array into individual elements and then merging them back in sorted order.

Input files

- [B, D]
- [A, C, E, F]
- [B, D]
- [A, C, E, F]
- [B, D]
- [A, C, E, F]
- [B, D]
- [A, C, E, F]

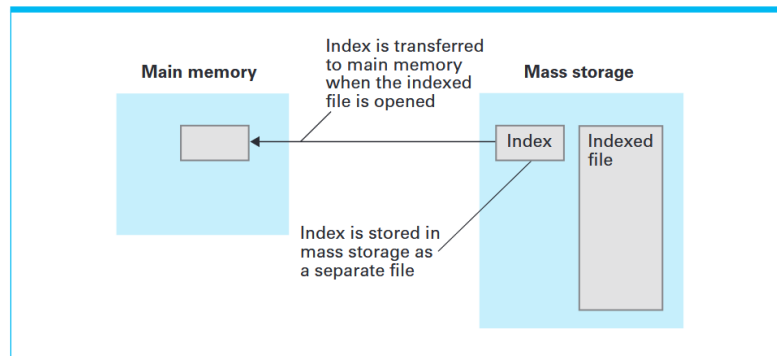
Output file

- [A]
- [A, B]
- [A, B, C]
- [A, B, C, D]
- [A, B, C, D, E]
- [A, B, C, D, E, F]

Indexed Files

An **indexed file** is a file system that uses an index to quickly locate records based on their keys.

Figure 9.17 Opening an indexed file



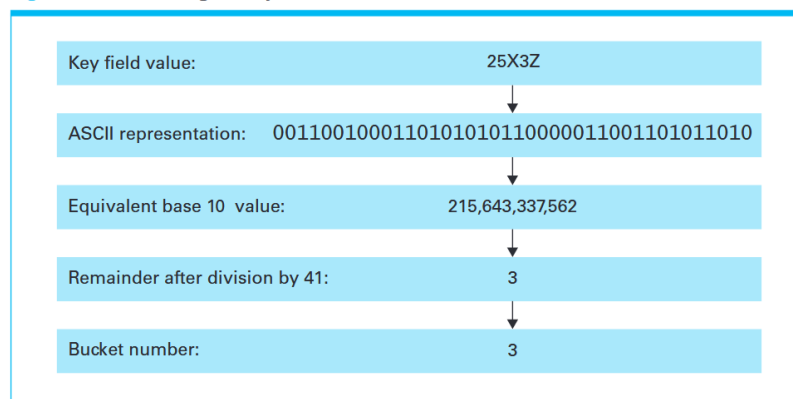
Hash Files

Hashing is a technique that uses a function to compute the storage location of a record directly from its key.

Buckets Storage space is divided into sections called buckets, each holding multiple records.
bucket = 存数据的“桶” 一个桶可以放多条记录

hash function converts a key into a bucket number.

Figure 9.18 Hashing the key field value 25X3Z to one of 41 buckets



When hashing is applied to a storage structure in mass storage, the result is called a **hash file**. When applied to a storage structure within main memory, the result is usually called a **hash table**.

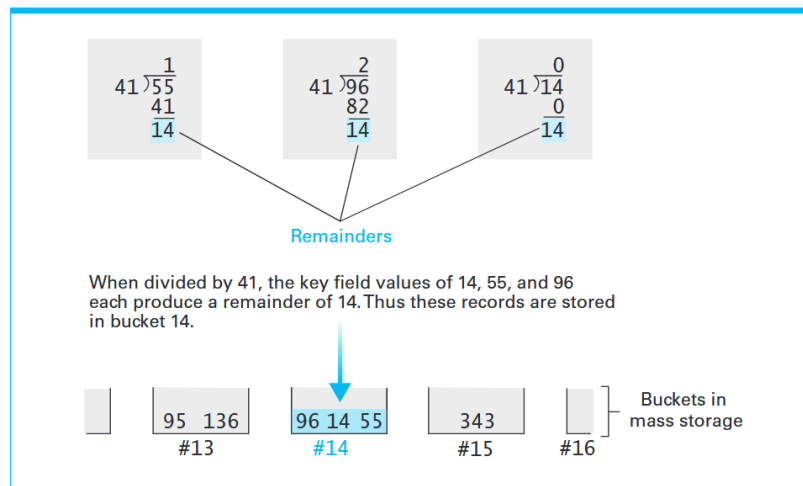
Clustering occurs when many records hash to the same bucket.

The **load factor** is the ratio of stored records to total capacity.

< 50% → 好

75% → 性能下降 (clustering 严重)

Figure 9.19 The rudiments of a hashing system



Authentication via Hashing

Hashing can be used to verify data integrity by comparing hash values.

发送方: 计算 hash

一起发送

接收方: 再算一次

比较

1. Follow the merge algorithm presented in Figure 9.15, assuming that one input file contains records with key field values equal to B and E while the other contains A, C, D, and F.

Input A: B, E

Input B: A, C, D, F

Output: A, B, C, D, E, F

2. The merge algorithm is the heart of a popular sort algorithm called the merge sort. Can you discover this algorithm? (Hint: Any nonempty file can be considered to be a collection of one-entry files.)

Merge sort works by dividing a file into smaller subfiles (each containing one record), then repeatedly merging them in sorted order to produce a fully sorted file.

3. Is being sequential a physical or conceptual property of a file?

Being sequential is a **conceptual property** of a file, since the file may be physically stored in different locations but is accessed in a logical sequence.

4. What are the steps required when retrieving a record from an indexed file?

Search for the key in the index

Use the index to find the location of the record

Retrieve the record from that location

5. Explain how a poorly chosen hash function can result in a hash storage system becoming little more than a sequential file.

A poor hash function causes clustering, so many records fall into the same bucket and must be searched sequentially.

6. Suppose a hash storage system is constructed using the division hash function as presented in the text but with six storage buckets. For each of the following key values, identify the bucket in which the record with that key is placed. What goes wrong and why?

a. 24 b. 30 c. 3 d. 18 e. 15
f. 21 g. 9 h. 39 i. 27 j. 0

six storage buckets $\rightarrow$ mod 6

key key mod 6 bucket

24	0	0
30	0	0
3	3	3
18	0	0
15	3	3
21	3	3
9	3	3
39	3	3
27	3	3
0	0	0

bucket 0: 24, 30, 18, 0

bucket 3: 3, 15, 21, 9, 39, 27

其他 bucket: 空

The divisor 6 shares common factors with the keys (such as 3), causing many keys to hash to the same buckets. This results in clustering and poor performance.

7. How many people must be gathered together before the odds(可能性) are that two members of the group will have birthdays on the same day of the year? How does this problem relate to the material in this section?

At least 23 people are required for the probability that two people share the same birthday to exceed 50%.

This relates to hashing because it shows that collisions are likely even when the number of possible values is large, similar to records mapping to the same bucket.

9.6 Data Mining

Data mining is the process of discovering previously unknown patterns(规律) in large collections of data.

A **data warehouse** is a static collection of data, often a snapshot of databases, used for data mining.

Types of Data Mining

-class description

Identifies properties that characterize a given class. 描述一个群体的特征

-class discrimination

Identifies properties that distinguish between two classes. 区分两个群体的差异

-cluster analysis

Discovers groups (clusters) within data. 自动发现数据分组 (原本不知道的类别)

-association analysis

Finds relationships between items. 找“经常一起出现”的东西

-Outlier analysis

Identifies data that deviates from the norm. 找“异常数据”

-sequential pattern analysis

Finds patterns over time. 分析“时间序列行为模式”

data cubes (理解型)

Data cubes represent data in multiple dimensions

1. Why is data mining not conducted on “online” databases?

they are continuously updated, making it difficult to identify stable patterns.

2. Give an additional example of a pattern that might be found by each of the types of data mining identified in the text.

-**class description** Identifying characteristics of customers who buy luxury products.

-**class discrimination** Distinguishing between customers who shop online and those who shop in stores.

-**cluster analysis** Grouping users into different age or interest groups automatically.

-**association analysis** Finding that customers who buy bread also buy milk.

-**Outlier analysis** Detecting unusual credit card transactions.

-**sequential pattern analysis** Identifying that customers buy a phone first, then accessories later.

3. Identify some different perspectives that a data cube might allow when mining sales data.

time, location, product type, customer age, and price.

4. How does data mining differ from traditional database inquiries?

Traditional database inquiries retrieve stored facts, whereas data mining discovers previously unknown patterns in data.

9.7 Social Impact of Database Technology

1. Should law enforcement agencies be given access to databases for the purpose of identifying individuals with criminal tendencies, even though the individuals might not have committed a crime?

Law enforcement agencies should have limited access to databases.

While such access can help prevent crime, it raises serious privacy concerns and may lead to unfair treatment of individuals who have not committed any crime.

2. Should insurance companies be given access to databases for the purpose of identifying individuals with potential medical problems, even though the individuals have not shown any symptoms?

Insurance companies should not have unrestricted access to such data.

Although it may improve risk management, it could lead to discrimination against individuals who are currently healthy.

3. Suppose you were financially comfortable. What benefits could you derive if this information were shared among a variety of institutions? What penalties could you suffer from the distribution of this same information? What if you were financially uncomfortable?

If I were financially comfortable, I could benefit from personalized services and better financial opportunities.

However, I might suffer from loss of privacy or misuse of my personal data.

If I were financially uncomfortable, the risks would be greater, as incorrect or biased (偏见) data could limit access to loans, jobs, or services.

4. What role does a free press(自媒体) have in controlling database abuse (滥用)? (For example, to what extent does the press affect public opinion or expose abuse?)

A free press helps control database abuse by exposing misuse of data and influencing public opinion, which can pressure organizations and governments to act responsibly.